

 perrierosas@gmail.com

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Artificial Intelligence undergraduate using combinatorial optimization and machine learning to build practical tools that solve real-world challenges.

Education

Nov 2023 – Present
Bénin

Bachelor of Science in Artificial Intelligence
IFRI, University of Abomey-Calavi
Expected completion: November 2026

Experience

- Jul 2026
Bénin

Applied Machine Learning Mentor
Sèmè City
 - Mentored the top 20 participants of the National Olympiad of Artificial Intelligence (NOAI) 2026, providing guidance in Machine Learning and supporting them throughout their technical challenges.
- Mar 2026 – Aug 2026
Bénin

Research Intern
LRSIA, University of Abomey-Calavi
 - Designed a MILP model for the MPVRP variant focusing on changeover costs (MPVRP-CC); initiated Constraint Programming modeling of the problem using successor and sequence variables.
 - Supervised the second-year AI undergraduate class project (Machine Learning course), contributing to the [ifri-mini-ml-lib](#) pedagogical library (neural network, NLP, and anomaly detection modules).
 - Contributed to the lab's internal projects and to technology monitoring.
 - Tools:** Gurobi, MaxiCP, MiniCP, Python, Java
 - Supervised by Dr. [Ratheil Houndji](#).
- Jun 2025 – Jan 2026
Bénin

AI Developer
IFRI & Tekbot Robotics
 - Developed and deployed an AI vision model and robotic controls for an automated waste-sorting system, allowing a robot arm to recognize and sort trash in real time to improve urban recycling.
 - Tools:** CNN, YOLOv5, PyTorch, Jetson Nano, CUDA, ONNX, TensorRT
 - Supervised by [David Akpovi](#).
- Jun 2025 – Aug 2025
South Africa

Machine Learning Engineer Intern
eTihuku
 - Built Review Scraper, an asynchronous API for collecting and aggregating customer reviews from Google Maps, HelloPeter, and X (Twitter), with background task processing and a scalable architecture.
 - Worked on SentiMaster, an AI platform that collects customer feedback from social media and phone calls, automatically translates and analyzes hidden sentiments, emotions and topics, and helps companies predict customer churn.
 - Tools:** FastAPI, Playwright, PyTorch, Transformers, BERT models, Airflow, Redis
 - Supervised by Dr. [Jeremiah Olamijuwon](#).

Selected Projects

- le-foncier-intelligent:** Developed a geospatial land-analysis pipeline combining multimodal OCR, geocoding, and satellite imagery to extract and assess information from topographic documents. Processed geospatial data using GeoPandas, Shapely and pyproj to support automated land-use analysis. [\[Code Repository\]](#)
- automated real estate mapping:** Designed an ETL pipeline to link address data, cadastral information, companies, and executives, in order to automatically generate enriched datasets for prospecting and analysis. [\[Code Repository\]](#)
- MDVRP:** Designed and implemented optimization algorithms (ALNS, PSO, Simulated Annealing, Linear Programming) for multi-depot vehicle routing (MDVRP) under capacity and distance constraints. [\[Code Repository\]](#)
- Opti'plan:** Co-built an automated scheduling model that reduced the university's thesis defense planning time from 3 days to under 5 minutes using Constraint programming.
- insurance-recommendation-system:** Developed an end-to-end machine learning pipeline for insurance product recommendation, including data processing, model training and comparison (XGBoost, CatBoost, Deep Learning), with a Flask API and web interface supporting prediction and automated model retraining. [\[Code Repository\]](#)
- ifri-mini-ml-lib:** Co-built an open-source Python library that reimplements classic AI algorithms from scratch for educational purposes; published on PyPI and used by IFRI students. [\[Code Repository\]](#)

Research & Presentations

Aug 2026 Lagos, Nigeria	Deep Learning Indaba A MILP Framework for Multi-Product Vehicle Routing Problem with Changeover Costs [Poster]
Nov 2025 Abomey-Calavi, Bénin	Benin Workshop on AI Evolutionary Algorithms: Principles, Variants, Application [Material]
Aug 2025 Abomey-Calavi, Bénin	ACP Summer School Constraint Programming for Sustainable Development [Certificate]
Aug 2025 Ouidah, Bénin	Young Girls' Leadership Camp Young Woman, Take the Lead in Shaping Your Future with AI [Link] Co-presented with Marie-Paule Lima
Oct 2024 Cotonou, Bénin	Future of AI Community Introduction to GPU Programming for AI [Material]

Skills

Programming	Python, Java, R, Bash
Machine Learning	PyTorch, JAX, Scikit-learn, Transformers
Optimization	Linear and Constraint Programming, Metaheuristics; Gurobi, MaxiCP
Tools	Git, Linux, Docker, FastAPI, GitHub Actions
Languages	French (native), English (B2)

Volunteering

Feb 2025 – Present	FRIARE Africa Participate in awareness initiatives on AI ethics, transparency and inclusion. [Link]
Jun 2025 – Aug 2026	THE LAB Co-led an AI awareness community in Bénin. [Link]
Jan 2025 – Jul 2026	IFRI AI & IoT Club Co-led the AI & IoT student club: delivered introductory sessions on Machine Learning and Optimization Algorithms, coordinated technical activities, and co-supervised AI projects [GitHub] .
Mar 2026 – Apr 2026	Applied Machine Learning Days (AMLDD) 2026 Co-organized the first edition of the Applied Machine Learning Days in Benin Republic. [Certificate]

Awards

2023 – 2026	Beninese Government Scholarship , awarded for academic excellence in the Baccalaureate .
2020 – 2026	Vallet Foundation Scholarship , awarded for academic excellence in the BEPC .
2025	2nd Place , Tekbot Robotics Challenge 2025. [Link]
2024	1st Prize , IndabaX Benin Republic Hackathon 2024. [Certificate]

Research Interests

Combinatorial optimization, industrial planning and scheduling problems, applied machine learning for social good.